

ETHERNAL FUND PROTOCOL

WHITEPAPER

Decentralized Personal Retirement Fund on Blockchain

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LEGAL NOTICE: This document is for informational purposes only and does not constitute an offer of securities, financial advice, or investment advice. The ETRF token is a utility token. Participation in the protocol involves significant risks. Consult a legal and financial advisor before making any decision.

Executive Summary

Eternal Fund Protocol is a decentralized personal retirement fund system built on blockchain (Arbitrum / Polygon), enabling anyone in the world to create and manage their own retirement fund autonomously, transparently, and without traditional financial intermediaries.

The protocol combines smart contracts written in Vyper with integrations to established DeFi protocols (Aave v3) and tokenized real-world assets (Ondo USDY / US T-Bills) to offer secure yields on USDC deposits. All business logic is encoded in auditable contracts, and funds always remain under the user's control.

Parameter	Value
Core contract	Vyper ^0.4.3 (PersonalFund, Factory, Treasury, Registry)
Deposit currency	USDC (6 decimals)
Management fee	5% per deposit (adjustable, maximum 10%)
Integrated yield	Aave v3 (DeFi) + Ondo USDY (RWA / T-Bills)
Fund timelock	15 to 50 years (user-configurable)
Utility token	ETRF — 1,000,000,000 total supply
Core contract license	BUSL-1.1 → GPL-2.0+ on 2030-12-31

Eternal's goal is to democratize access to retirement savings, offering a transparent and non-custodial alternative to traditional pension systems, accessible globally with just a wallet and stablecoins.

1. Problem and Solution

1.1 The Global Retirement Problem

Traditional pension systems face a structural crisis worldwide: pension funds with growing deficits, inflation eroding purchasing power, and increasing labor informality leaving millions of workers without pension coverage. In Latin America, less than 40% of workers actively contribute to formal pension systems.

Conventional savings instruments (fixed deposits, mutual funds) present significant barriers: high minimums, opaque fees, dependence on intermediaries, and geographically restricted access.

1.2 The DeFi Opportunity

Blockchain technology and DeFi protocols enable, for the first time, long-term financial instruments that are:

- **Non-custodial:** users maintain control of their funds at all times.
- **Transparent:** all logic is encoded in public, verifiable contracts.
- **Borderless:** accessible from anywhere with an internet connection.

- **Programmable:** saving, investment, and withdrawal rules execute automatically.
- **Composable:** integrate established DeFi protocols to generate yield.

1.3 The Solution: Eternal Fund Protocol

Eternal Fund Protocol resolves these limitations through smart contracts that allow each user to create their own personal retirement fund on blockchain, with customized parameters (retirement age, monthly deposit, risk tolerance, preferred investment protocol) without ceding control of their assets to any third party.

2. Protocol Architecture

The protocol consists of interdependent smart contracts organized in functional layers. All core contracts are written in Vyper ^0.4.3 for its superior security and auditability compared to Solidity.

2.1 Core Contracts (BUSL-1.1)

PersonalFund.vy — The central contract. Each user has their own deployed instance (minimal proxy EIP-1167). Manages the complete fund lifecycle: initial deposits, monthly deposits, DeFi investments, retirement withdrawal, and emergency withdrawal.

PersonalFundFactory.vy — Deploys new PersonalFund instances using the minimal proxy pattern (EIP-1167 / clones), significantly reducing gas costs. Validates all input parameters before deployment.

Treasury.vy — Manages protocol fee collection and acts as arbiter for early withdrawal requests. Default fee: 5% (500 bps). Maximum fee: 10% (1000 bps).

ProtocolRegistry.vy — Curated catalog of DeFi protocols available for investment, with metadata: name, category, estimated APY, risk level, TVL, and verification status.

UserPreferences.vy — Manages individual user configuration: selected DeFi protocol, auto-compound, risk tolerance, and deposit routing strategy.

2.2 DeFi Adapters (BUSL-1.1)

AaveV3Adapter.vy — Bridge to the Aave v3 lending market on Arbitrum One. Implements safe allowance pattern (reset to 0 before each approval). Pool: 0x794a61358D6845594F94dc1DB02A252b5b4814aD.

OndoUSDYAdapter.vy — Adapter for investing in USDY, a tokenized promissory note backed by short-term US Treasury bills. Historical APY: 4.5%–5.5%. Uses secondary market route via Uniswap v3 (USDC → USDY). Estimated slippage: 0.01%–0.05%.

3. Personal Fund Lifecycle

3.1 Fund Creation

Parameter	Description
principal	Initial seed capital in USDC (0 – 100,000 USDC)
monthlyDeposit	Fixed amount to deposit every 30 days
currentAge	User's current age (validated by factory ranges)
retirementAge	Target retirement age (> currentAge)
desiredMonthly	Desired monthly amount during retirement
yearsPayments	Years of payments in the retirement phase (1–50)
timelockYears	Years until retirement / fund lockup period (15–50)
selectedProtocol	Initial DeFi adapter address (must be approved)

3.2 Accumulation Phase

During the lockup period (`timelockYears`), the user makes periodic monthly deposits and accumulates yield from DeFi protocols. The protocol records missed months (`missedMonths`) informatively without applying penalties. Extra deposits have a 24-hour window (`EXTRA_GRACE_PERIOD`) to be rescinded with a 1% penalty.

3.3 Early Withdrawal

If the user needs access to their funds before the timelock expires, they must request approval from the protocol Treasury via `requestEarlyRetirement(reason)`. The admin can approve or reject the request.

3.4 Emergency Withdrawal

In extreme situations, the admin can announce an emergency withdrawal via `announceEmergency()`. There is a mandatory 48-hour waiting period (`EMERGENCY_DELAY = 172,800` seconds) before the user can execute `emergencyWithdraw()`.

3.5 Retirement Phase

Upon completion of the timelock period, the owner activates the retirement phase with `retirement()`. The system automatically configures monthly auto-withdrawals for the desired amount (`desiredMonthly`). Minimum interval between withdrawals: 7 days (604,800 seconds).

4. ETRF Token — Eternal Fund Token

UTILITY DECLARATION: ETRF is exclusively a utility token. It does NOT represent equity, debt, dividends, or any expectation of profit derived from the work of third parties. It is not designed as an investment instrument or security under any jurisdiction.

4.1 Token Use Cases

- **Premium Mode:** users holding a minimum of ETRF unlock advanced application features (reports, investment strategies, alerts). The token is only verified, not consumed.
- **Fee Discount:** users can pay management fees in ETRF instead of USDC, receiving a discount. ETRF tokens used are burned, reducing the circulating supply.
- **Prepaid Credits:** users acquire ETRF in advance and the protocol gradually burns them as usage fees are generated.

4.2 Tokenomics

Total supply: 1,000,000,000 ETRF (fixed, non-mintable post-deploy).

Allocation	%	Tokens (ETRF)	Details
Public Sale	30%	300,000,000	Rounds via <code>SaleETRF</code> → <code>VestingETRF</code>
Team	20%	200,000,000	1-year cliff + 25% unlock + 75% linear over 3 years
Ecosystem	20%	200,000,000	Incentives, partnerships, integrations
Treasury	15%	150,000,000	Receives USDC from sales and unsold tokens
Community	10%	100,000,000	Community programs, rewards
Reserve	5%	50,000,000	Strategic reserve

4.3 Vesting Mechanism

Public sale tokens are distributed linearly after a round-configurable cliff period. Team vesting: 1-year cliff → 25% instant unlock → 75% linear over the following 3 years.

5. Security and Risk Management

5.1 Why Vyper

All core contracts are written in Vyper, a smart contract language designed with an emphasis on security and auditability. It prohibits multiple inheritance, operator overloading, recursion, and inline modifiers, significantly reducing attack surface.

5.2 Access Controls

Control	Mechanism
Fund owner	Only the PersonalFund owner can make deposits and withdrawals
Protocol admin	Separate role for administrative operations (Treasury, Factory, Registry)
Protocol whitelist	Only active and verified protocols in ProtocolRegistry are usable
Fund registry	Only funds registered in Factory can report fees to Treasury

5.3 Risks to Consider

The following risks are inherent to DeFi protocols and blockchain usage. Users must understand them before participating.

- **Smart contract risk:** no code is 100% bug-free. Funds may be lost in the event of an exploit.
- **DeFi protocol risk:** Aave v3 and Ondo USDY are external. An exploit there could affect deposited funds.
- **Stablecoin depeg risk:** USDC may lose its USD parity under extreme circumstances.
- **Regulatory risk:** the legal framework for digital assets is uncertain and varies by jurisdiction.
- **Liquidity risk:** in market stress, liquidity in Uniswap v3 USDY/USDC pools could decrease.
- **Admin risk:** dependence on the Eternal Foundation for key administrative operations.

6. Legal and Regulatory Framework

6.1 Contract Licensing

Contract	License	Term
PersonalFund.vy	BUSL-1.1	Until 2030-12-31 → GPL-2.0+
PersonalFundFactory.vy	BUSL-1.1	Until 2030-12-31 → GPL-2.0+
Treasury.vy	BUSL-1.1	Until 2030-12-31 → GPL-2.0+
AaveV3Adapter.vy	BUSL-1.1	Until 2030-12-31 → GPL-2.0+
OndoUSDYAdapter.vy	BUSL-1.1	Until 2030-12-31 → GPL-2.0+
ProtocolRegistry.vy	Apache-2.0	Perpetual
UserPreferences.vy	Apache-2.0	Perpetual
DateTime.vy	Apache-2.0	Perpetual
ETRF.vy	MIT	Perpetual
SaleETRF.vy	MIT	Perpetual
VestingETRF.vy	MIT	Perpetual

6.2 Token Classification

The ETRF token has been designed as a pure utility token. Its value derives solely from its use within the Eternal Fund ecosystem and not from any expectation of profit. Each user is responsible for determining the legal and tax classification of the ETRF token in their jurisdiction.

6.3 KYC / AML — Compliance by Design

The protocol operates in a non-custodial manner and does not require user identification at the smart contract level. The OndoUSDYAdapter uses the Uniswap v3 secondary market route to avoid direct KYC with Ondo Finance. Any

change in Ondo's access policies could affect this integration.

Timelock as a structural AML mitigator. The protocol's timelock mechanism (15 to 50 years) incorporates an anti-money-laundering compliance property by architectural design. Funds deposited in PersonalFund are locked for the entire retirement period, with no early withdrawal possible without explicit Treasury approval. This makes the retirement fund product structurally incompatible with the classic money laundering cycle (placement → layering → integration): the integration stage — where the actor recovers funds with a legitimate appearance — would require waiting decades, effectively eliminating any economic incentive to use the protocol for illicit purposes.

This property is recognized in compliance literature as *friction by design*: unlike reactive KYC solutions that identify bad actors post-hoc, the timelock removes the protocol's attractiveness as a laundering vehicle before the transaction occurs. Early withdrawal requires discretionary approval from the Foundation, adding an additional layer of human oversight for exceptional cases.

Scope note: this compliance by design property applies specifically to the retirement fund product (PersonalFund). The ETRF token, whose vesting period is measured in months, is subject to standard KYC/AML terms described in the Token Sale Terms and Conditions. Notwithstanding the architectural design described above, the protocol may be subject to additional regulatory obligations in different jurisdictions. The Eternal Foundation reserves the right to implement complementary compliance mechanisms as required by applicable law.

7. Roadmap

Phase	Milestones
Q2 2026 — Testnet	Full deployment on Arbitrum Sepolia. Security audit. Bug bounty program.
Q3 2026 — Mainnet v1	Launch on Arbitrum One. ETRF public sale (Seed Round). First user funds.
Q4 2026 — Expansion	Integration of new DeFi protocols. Deployment on Polygon. More RWA adapters.
2027 — Decentralization	On-chain governance for Protocol Registry. Reduction of admin privileges.
2030-12-31 — Open Source	Migration from BUSL-1.1 to GPL-2.0+ licenses. Fully open-source protocol.

8. Team and Contact

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9. Legal Disclaimer

This whitepaper has been prepared solely for informational purposes about the Eternal Fund Protocol and the ETRF token. It does not constitute, nor should it be interpreted as, an offer or solicitation to purchase or subscribe to securities, financial assets, or other instruments in any jurisdiction.

Investment in cryptocurrencies and DeFi protocols entails substantial risks, including the possibility of total loss of invested capital. The Eternal Foundation accepts no liability for direct, indirect, incidental, or consequential losses.

This document may contain forward-looking statements based on the team's current expectations. The roadmap is subject to change without notice.